List of Seminar Topics

Last update: Oct 11, 2026

Seminar Umbrella Topic

- **General: Papers in the fields of systems for data engineering, data management, and machine learning**
- **This Semester's Umbrella Topic: "Robust and Adaptive Query Processing"**
  - **Motivation**
    - Traditionally, database query processing is divided into an optimization phase, which determines an optimal plan for the query, and an execution phase, which executes this plan
    - During optimization, different logically equivalent plans are enumerated and the plan with the lowest cost with respect to some cost model is chosen
    - Cost estimation is largely based on estimates of the cardinalities (sizes) of intermediate results
    - Unfortunately, these **estimates are often quite wrong** resulting in **bad query execution plans** that may take orders of magnitude longer to execute than the optimal plan
    - Moreover, **additional unknowns** further complicate the efficient query processing, e.g., unknown properties of base data and input datasets, the access to external data sources, query parameters, and the system utilization at run-time

Seminar Umbrella Topic

- **General: Papers in the fields of systems for data engineering, data management, and machine learning**
- **This Semester's Umbrella Topic: "Robust and Adaptive Query Processing"**
  - **How can we ensure efficient query processing in the presence of so many unknowns?**
  - **Ideas to address these challenges**
    - **Improved creation/management of statistics**: Try to make more accurate information on the base data and intermediate results available during query optimization
    - **Robust query optimization**: Rather than searching for an optimal query execution plan, search for a robust plan that is insensitive to inaccurate estimates
    - **Adaptive query processing**: Introduce a feedback loop to exploit information gathered at query run-time for adapting the query execution dynamically
  - **Active field of research**
    - After an **initial peak in the 2000s**, robust and adaptive query processing is **still an active research area today**
    - The seminar collects a **broad range of relevant papers** addressing various aspects of robust and adaptive query processing

Seminar Topics (1/4)

- **Improved Creation/Management of Statistics**
  1) Stillger et al.: **LEO - DB2's LEarning Optimizer** (VLDB 2001) [link]
  2) Chaudhuri and Narasayya: **Automating Statistics Management for Query Optimizers** (Trans Knowl Data Eng 2001) [link]
  3) Wu et al.: **Sampling-Based Query Re-Optimization** (SIGMOD 2016) [link]
  4) Dutt et al.: **Selectivity Estimation for Range Predicates using Lightweight Models** (PVLDB 2019) [link]

Seminar Topics (2/4)

- **Robust Query Optimization**
  5) Babcock and Chaudhuri: **Towards a Robust Query Optimizer: A Principled and Practical Approach** (SIGMOD 2005) [link]
  6) Babu et al.: **Proactive Re-optimization** (SIGMOD 2005) [link]
  7) Doraiswamy et al.: **On the Production of Anorexic Plan Diagrams** (VLDB 2007) [link]
  8) Bizarro et al.: **Progressive Parametric Query Optimization** (Trans Knowl Data Eng 2009) [link]
  9) Dutt and Haritsa: **Plan Bouquets: Query Processing without Selectivity Estimation** (SIGMOD 2014) [link]
  10) Wei and Trummer: **ROME: Robust Query Optimization via Parallel Multi-Plan Execution** (SIGMOD 2024) [link]

Seminar Topics (3/4)

- **Adaptive Query Processing 1990s/2000s**
  11) Kabra and DeWitt: **Efficient Mid-Query Re-Optimization of Sub-Optimal Query Execution Plans** (SIGMOD 1998) [link]
  12) Urhan et al.: **Cost Based Query Scrambling for Initial Delays** (SIGMOD 1998) [link]
  13) Avnur and Hellerstein: **Eddies: Continuously Adaptive Query Processing** (SIGMOD 2000) [link]
  14) Markl et al.: **Robust Query Processing through Progressive Optimization** (SIGMOD 2004) [link]
  15) Ives et al.: **Adapting to Source Properties in Processing Data Integration Queries** (SIGMOD 2004) [link]
  16) Babu et al.: **Adaptive Ordering of Pipelined Stream Filters** (SIGMOD 2004) [link]
  17) Li et al.: **Adaptively Reordering Joins during Query Execution** (ICDE 2007) [link]

Seminar Topics (4/4)

- **Adaptive Query Processing 2010s/2020s**
  - 18) Raducanu et al.: **Micro Adaptivity in Vectorwise** (SIGMOD 2013) [link]
  - 19) Karnagel et al.: **Adaptive Work Placement for Query Processing on Heterogeneous Computing Resources** (PVLDB 2017) [link]
  - 20) Wolf et al.: **On the Calculation of Optimality Ranges for Relational Query Execution Plans** (SIGMOD 2018) [link]
  - 21) Kissinger et al.: **Adaptive Energy-Control for In-Memory Database Systems** (SIGMOD 2018) [link]
  - 22) Kohn et al.: **Adaptive Execution of Compiled Queries** (ICDE 2018) [link]
  - 23) Trummer et al.: **SkinnerDB: Regret-Bounded Query Evaluation via Reinforcement Learning** (SIGMOD 2019) [link]
  - 24) Menon et al.: **Permutable Compiled Queries: Dynamically Adapting Compiled Queries without Recompiling** (PVLDB 2020) [link]
  - 25) Justen et al.: **POLAR: Adaptive and Non-invasive Join Order Selection via Plans of Least Resistance** (PVLDB 2024) [link]